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IT3182 Essentials of Artificial Intelligence Group Project
Final Report

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Purpose and implementation baseline

DIAR analyses professional evidence from resumes, public GitHub profiles and supplied LinkedIn text. This report describes the implemented system, its AI methods, evaluation and privacy safeguards. The implementation reviewed is GitHub main commit 59662cd, checked on 17 October 2026, which incorporates implementation commit f635bd6 through merge commit e2e0440 and later interface and documentation commits. All nine agreed improvements are implemented in this baseline.

The prototype supports professional self-assessment through a consolidated profile, project-defined benchmarks and explained recommendations. Protected export, recommendation navigation and reproducible evaluation complete the current workflow; the backend keeps saved-report history, although the history panel is hidden at 59662cd. The system remains a local educational prototype with explicit evidence and coverage limits.

Submission context: final university group project report for Group 41. Individual contributions are recorded in Appendix A (Individual contributions) of this report. Evaluation results are dated records from the repository and are distinguished from independent user validation.

Abstract and report contents

Abstract

DIAR addresses fragmented professional information across resumes, public GitHub profiles and supplied LinkedIn text. We combine dictionary-based language processing, structured benchmark knowledge, supervised role classification, fuzzy evidence summaries and rule-based reasoning to produce prioritised recommendations. The report retains the source context needed to distinguish information that was not supplied from supported claims. Visibility preferences guide professional-sharing advice, while an independent policy masks saved and returned information. The local interface supports protected JSON export, printing and deletion of the displayed report and recommendation filters, while the backend keeps report history.

The completed improvements include input validation, conservative experience handling, skill-context extraction, ML evidence safeguards, request recovery and optional bounded review of public repository files. Findings include masked details and checked/skipped coverage. The latest recorded regression run passed 199 tests. Evaluation on 75 labelled profiles obtained 98.67% mean accuracy across five shuffled stratified folds and 100% on a 15-example holdout. These small-dataset results do not establish real-profile accuracy or calibrated probabilities. The prototype demonstrates explainable self-assessment; competence verification, account access control and broader external validation remain outside the completed scope.

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1 Introduction and problem background

Introduction

Students, job seekers and early-career professionals often describe their experience differently across a resume, GitHub and LinkedIn. A resume summarises qualifications and employment, GitHub can provide public project evidence, and LinkedIn text can express interests and professional direction. Reviewing these sources separately makes omissions and inconsistencies difficult to identify. A person may also want to demonstrate professional ability without publishing unnecessary personal details.

Problem background

The existing manual approach requires users to read several profiles, compare terminology and decide which actions matter most. Keyword lists alone can misinterpret “I plan to learn Docker” as established experience, or “no experience with Java” as a positive Java claim. Missing input may be confused with absent ability. Public contact information can occur in repository documentation as well as profile metadata. These problems motivate a system that records where evidence came from, explains uncertainty and connects each recommendation to a clear reason.

Problem statement

How can professional information from several user-selected sources be consolidated and compared with a target role to generate explainable development and privacy recommendations, while distinguishing unavailable information from observed evidence and respecting the user’s sharing preferences?

Target users and scope

The target users are university students and professionals reviewing their own digital identity. The user chooses AI Engineer, Software Engineer, Data Scientist, Researcher or Entrepreneur as the benchmark. These are project-defined reference profiles, not independently validated hiring standards. The tool assesses supplied information and returned public GitHub evidence; it cannot infer a complete professional record from an incomplete submission.

The delivered architecture is a local Python and FastAPI service, SQLite storage and a plain JavaScript interface. Text-based PDF resumes and pasted LinkedIn text are supported. Public GitHub retrieval requires a network connection; text analysis and prepared local models can operate without external model services. LinkedIn scraping, OCR, private repositories, historical repository content and automatic edits to external accounts are outside the assessed scope. [1, 2]

2 Objectives and selected AI techniques

Project objectives

Our objectives are to extract and normalise professional information, consolidate source-aware evidence, compare it with a selected benchmark, provide a supplementary predicted role and explain prioritised recommendations. The delivered system also validates inputs, preserves useful results after failures, reviews selected public text files for privacy patterns and lets users filter, export, print and delete protected reports.

Concept	Use in DIAR	Why it is suitable
Language processing	Dictionary aliases, regular expressions, context categories and typo matching	Makes extraction inspectable and practical for a small local prototype.
Knowledge representation	JSON benchmark identities and structured evidence fields	Expresses required/preferred skills, project keywords and threshold rules explicitly.
Supervised classification	TF-IDF with logistic regression over five role labels	Provides a lightweight learned comparison alongside hand-authored rules.
Fuzzy logic	Triangular memberships map ratios to linguistic labels	Represents gradual skill matching and activity rather than only yes/no outcomes.
Rule-based reasoning	Conditions produce actions, priorities and explanations	Allows a user to trace advice to evidence and benchmark requirements.
Priority queue ordering	Heap ranks fired actions and removes duplicates	Provides transparent ordering of available actions with stable ties.

Justification and proposal changes

DIAR applies more than the three AI concepts required by the assignment. Each technique has a distinct role: extraction structures text, classification estimates a role, fuzzy memberships summarise evidence, and rules produce advice. The queue orders that advice. It is a ranking procedure over candidate actions, not a multi-step planning algorithm such as A* or an optimal career plan.

The proposal named React, spaCy or Transformers and scikit-fuzzy. The delivered prototype uses vanilla JavaScript, dictionary and regular-expression extraction, and a custom fuzzy implementation. A separate scikit-learn classifier provides learned role prediction. These implementation choices reduce setup complexity while preserving the proposal's core extraction, benchmarking and explainable recommendation workflow. [1, 3, 4, 7]

3 AI scoring and decision process

Skill evidence and consolidation

Skill aliases map to canonical names. Extraction preserves line and clause boundaries, assigns claimed, uncertain, planned or negated context, and avoids counting overlapping aliases as separate occurrences. Typo matching uses difflib with a similarity cutoff of 0.82 for eligible single words. These are heuristic language rules: a claimed skill is a textual assertion rather than independently verified competence.

Consolidation gives claimed evidence precedence over uncertain, planned and negated mentions across sources. GitHub language evidence enters the claimed pool. Conflicts therefore need user review. Frequency weights use $w = 1 + 0.4 \ln(\min(\text{count}, 5))$ for counts above one, otherwise $w = 1$. This caps the influence of repeated keywords.

Contextual score and fuzzy summary

Required skills have base weight 1 and preferred skills base weight 0.5. Claimed evidence receives full credit, uncertain evidence half credit and planned evidence one-quarter credit. The contextual score is the credited weighted total divided by the total weighted benchmark points, bounded to [0, 1]. This is a project-defined evidence score; the learning-intent credit is not proof of competence.

The separate fuzzy skill ratio is 0.7 times required-skill coverage plus 0.3 times preferred-skill coverage, using claimed and uncertain skills as active evidence. Low, moderate and strong memberships use triangular parameters (-0.2, 0, 0.5), (0.15, 0.5, 0.85) and (0.5, 1, 1.2). The largest membership determines the label. Activity uses the ratio of recently active repositories to retrieved portfolio repositories. Completeness uses source flags and is interpreted as supplied coverage. These ratios are not calibrated employment-success probabilities.

Rules and explanation

Required skills absent from supplied evidence generate high-priority actions when usable skill evidence exists. Explicitly negated skills can generate structured training; planned skills can generate learning continuation or acceleration. Experience advice consults experience status, and repository advice requires adequate retrieval coverage. Privacy rules use source-specific masked findings. Every recommendation records its rank, priority, rule identifier and reason, with evidence links where available.

The queue assigns high = 3, medium = 2 and low = 1, then breaks equal-priority ties by insertion order. It deduplicates identical actions. The implemented ranking does not add a separate benchmark-relevance weight. Rule explanations describe the trigger for advice. ML explanations separately list matching vocabulary ranked by TF-IDF weight, with alphabetical ties; they do not claim to measure class contributions. [1]

4 System design

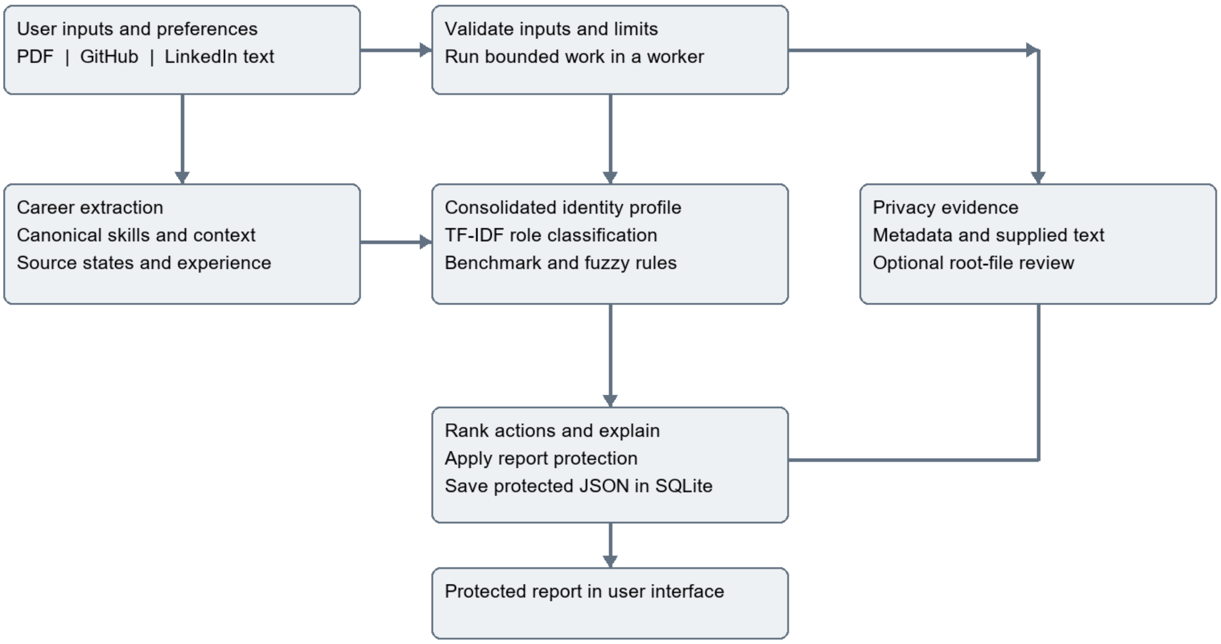

User inputs and preferences
PDF | GitHub | LinkedIn text
Validate inputs and limits
Run bounded work in a worker
Career extraction
Canonical skills and context
Source states and experience
Consolidated identity profile
TF-IDF role classification
Benchmark and fuzzy rules
Privacy evidence
Metadata and supplied text
Optional root-file review
Rank actions and explain
Apply report protection
Save protected JSON in SQLite
Protected report in user interface

Figure 1 Current DIAR processing and storage flow. Repository-file text feeds privacy evidence only; it is excluded from career extraction and ML inputs.

Input and output boundaries

The analysis request includes optional resume, GitHub username and LinkedIn text, plus benchmark identity and visibility level. Independent controls specify whether the resume is publicly shared, the report redaction policy and whether repository-file review is enabled. The output contains the consolidated profile, benchmark comparison, ML prediction, gap analysis, visibility assessment, recommendations, explanation and report metadata.

Persistence and API

Endpoint	Purpose
POST /api/analyze	Validate, construct, protect and save a new report.
GET /api/benchmarks	Return the five benchmark options.
GET /api/visibility-levels	Return supported visibility choices.
GET /api/reports	Return up to 50 recent protected report summaries.
GET /api/reports/{id}	Read a report under its stored protection policy.
DELETE /api/reports/{id}	Remove the selected saved database row.
GET /api/health	Check service responsiveness.

SQLite stores report ID, benchmark, visibility, protected GitHub username, JSON report and timestamp. Historical records are read under their stored policy or a legacy fallback; they are not automatically rescored. The local prototype has no authentication or per-user ownership controls. [1]

5 Implementation details

Technology and module responsibilities

Component	Implementation
Interface	frontend/index.html and report-review.js; plain HTML, CSS and JavaScript with no npm build.
Request handling	backend/app/main.py; FastAPI forms, upload limits and worker orchestration.
Extraction	modules/extraction.py; PyMuPDF, dictionary aliases, regex and public GitHub requests.
Profile construction	modules/identity_construction.py; merge evidence, context and source states.
Benchmarks and ML	data/benchmarks.json; ml_classifier.py; TF-IDF and logistic regression.
Reasoning and ranking	alignment_engine.py, fuzzy_logic.py and recommendation_engine.py.
Privacy and explanation	privacy.py, privacy_assessment.py, repository_privacy.py and explainable_ai.py.
Persistence	database.py; SQLAlchemy sessions and local SQLite report storage.

Input validation and evidence handling

At least one source must be supplied. A resume is limited to 5 MiB, 30 pages and 100,000 extracted characters; LinkedIn text is limited to 100,000 characters. Malformed, encrypted and unreadable PDFs receive explanatory errors. No OCR is performed. A failed GitHub-only request returns 422 rather than saving an empty profile. Invalid uploads reject the request even if another source is supplied, allowing the user to correct or remove the file.

Source states distinguish not supplied, analysed, partial and failed. Experience records a value, status and basis. Unknown experience uses null in the new structure; the old numeric field can remain zero for compatibility. Labelled experience date ranges are merged to avoid overlap, education dates are excluded, and the maximum supported estimate across resume and LinkedIn is used instead of adding duplicate totals.

Recovery and local execution

Blocking PDF, network and model work runs in an AnyIO worker. The backend has a 40-second report-construction deadline after bounded upload reading, and the browser stops waiting after 45 seconds. A backend deadline response skips report persistence; browser cancellation alone does not ensure the server stopped. The previous successful report remains visible during another request and after failure. A model exception preserves the rule-based result.

The supported setup uses Python 3.12.10 and pinned runtime/development dependencies, with a full resolved requirements-lock.txt. A fresh environment was documented in the development verification record at e2e0440; that record was removed from main in 6d7678f, and the GitHub Actions workflow now provides the hosted reproducibility check. The backend runs with python -m uvicorn app.main:app --reload --port 8000; the static frontend can be served on port 5173. GitHub retrieval needs network access. Dictionary processing and the local classifier do not call an external AI service. [1, 5]

6 Final product improvements and status

All nine improvements are implemented in the reviewed code. IDs 1 to 3 were integrated through PR #6, ID 7 through PR #7, ID 9 through PR #8, ID 4 through PR #9, IDs 5 and 6 through PR #10, and ID 8 through PR #11. GitHub main was rechecked at 59662cd on 17 October 2026. The later commits de4d93f, 6d7678f, 186ef77 and 59662cd change the interface, README and documentation but not the backend: they remove the suggested-first-steps shortcuts, hide the saved-report history panel and delete the docs folder apart from the restored model-evaluation.json. The status below reflects the implementation and merge history at that commit. [1]

ID	Improvement	Assessed status
1	Input and upload validation	Implemented
2	Insufficient evidence and unknown experience	Implemented
3	Skill context and alias deduplication	Implemented
4	Trustworthy ML evidence and keyword explanations	Implemented
5	Saved-report history and protected export	Implemented; history panel hidden in 59662cd
6	Recommendation filters and action links	Implemented; first-step shortcuts removed in de4d93f
7	Failure recovery and responsiveness	Implemented
8	Reproducible setup, CI and dataset records	Implemented
9	Bounded public repository file privacy scan	Implemented

Public repository file review

The optional scan is off by default and operates on fetched public repositories, including forks. It selects up to five repositories, one supported root README and root CONTRIBUTING.md and SECURITY.md. Supported README formats include Markdown, reStructuredText, plain text and an extensionless README. This initial allowlist is the meaning of selected text files; arbitrary user-chosen paths and recursive discovery are not implemented.

Limits are three files per repository, 100 KiB decoded text per file, 1 MiB decoded text overall, 25 extra requests, 256 KiB per transport response, a five-second request ceiling and a 20-second elapsed scan budget within the overall deadline. Fixed GitHub API paths, bounded streaming and recorded revisions restrict retrieval. Private repositories, binaries, symlinks, submodules, nested source files, linked pages, issues and Git history are excluded.

The scanner detects supported email, possible phone, explicitly labelled street address and date-of-birth patterns. Findings show masked excerpts, repository/path labels, line numbers and revision context. Coverage records checked/skipped files and reasons for partial or unavailable results. Raw downloaded text is not saved, even under the least restrictive report policy, and does not enter career scores or predictions. Source links are deliberately omitted where they could defeat masking. [1, 6]

7 Report review and reproducible setup

Improvement 4 ML evidence and explanations

Empty text and text with no trained vocabulary now return an insufficient-evidence state without a role distribution. Model load, training or inference failures return unavailable ML while retaining rule analysis. Matching unigram and bigram vocabulary is sorted by TF-IDF weight and then alphabetically. Probabilities are explicitly labelled as uncalibrated model outputs. No evaluated low-probability threshold exists, so mixed or weak text with vocabulary matches may still receive a role ranking.

Improvement 5 Saved report history and export

Commit 59662cd hides the Saved reports panel with a display:none style, so the interface no longer lists or reopens earlier reports. The backend still returns the latest 50 protected summaries through GET /api/reports and individual reports through GET /api/reports/{id}, and the hidden panel's refresh and reopening code remains in report-review.js. The displayed saved report offers Export JSON, Print report / Save PDF and confirmation-based deletion. JSON export uses the protected report currently displayed and an ID-based filename. Browser print styling includes the complete recommendation list and restores screen filters afterwards. Deletion does not remove downloaded copies.

Improvement 6 Recommendation navigation

Category and priority filters preserve original ranks and support career, privacy and legacy or uncategorised actions. The suggested-first-steps list and public repository-file action shortcuts were removed from the navigation panel in de4d93f. Privacy evidence cards keep View recommended action links, which reveal recommendations hidden by active filters. Historical actions remain readable without silent reclassification. The interface changes navigation rather than introducing a new risk score or ranking policy.

Improvement 8 Reproducible setup and verification

Direct dependencies are pinned and a full lockfile records a fresh Python 3.12.10 environment. The test fixtures use disposable SQLite stores, block real HTTP and keep generated model artefacts in temporary storage. The GitHub Actions workflow installs dependencies, checks compatibility, runs regression tests and performs offline evaluation. The evaluation script records dataset hash, class counts, split indices, duplicate screening and metrics. It updates the local model only when --save-model is supplied.

Verification boundaries

Implementation status does not establish that every environment or user flow has been validated. The fresh-environment run and browser checks recorded at e2e0440 support the completed features, but native browser print/PDF output was not visually verified in those checks, and they predate the de4d93f interface rewrite. The GitHub Actions Backend regression workflow completed successfully for e2e0440 and for every later main commit through 59662cd. For this report revision the test suite was also re-run on 59662cd: 199 passed with three deprecation warnings under Python 3.13.5; CI uses Python 3.12.10. [1]

8 Testing and sample results

Regression verification

The latest recorded backend regression run was completed on 11 October 2026 in a fresh Python 3.12.10 environment: 199 tests passed, with three existing FastAPI/Starlette deprecation warnings. The command was python -m pytest tests -q from backend. Fixtures use synthetic inputs, mocked HTTP and isolated databases. Recorded dependency compatibility checks and startup health/benchmark requests also passed. These results come from the development verification record at e2e0440. On 17 October 2026 the same command was re-run on main commit 59662cd: 199 tests passed with three deprecation warnings under Python 3.13.5, while CI runs the suite on Python 3.12.10.

Scenario	Input or condition	Observed or tested result
Validation	Empty/whitespace sources; malformed username or PDF	Rejected; invalid analysis does not save a misleading report.
Skill context	No experience with Java. New line: Python and SQL. I plan to learn Docker.	Python/SQL claimed; Java negated; Docker planned; each frequency is one.
Weak evidence	Contact email only	Skill-gap judgement withheld; zero model vocabulary yields ML insufficient evidence.
Experience	Education 2020 to 2024; Python and SQL	Experience unknown; education dates do not become work years.
Recovery	Worker deadline or model failure	Deadline saves no report; failed ML retains rule analysis.
File privacy	Mock public root files with contact patterns and partial failures	Masked line/path evidence, explicit coverage and bounded retrieval; career/ML inputs unchanged.
Protected storage	Create/read under each protection policy	Saved and returned payloads obey policy; raw scanned files are excluded.

Worked example of alignment

A synthetic extraction/alignment example uses the text "No experience with Java. Python and SQL. I plan to learn Docker." with the Software Engineer benchmark and Privacy Focused visibility. The earlier recorded direct execution found Python and SQL claimed, Java negated, Docker planned and experience unknown. Its fuzzy ratio was 0.200/low and contextual score 0.193. The relevant rules were R1-negated-java and R2-planned-docker. Skills absent from this short input remain gaps in supplied evidence rather than verified lack of ability.

Verification boundaries

Browser checks recorded at e2e0440, before the docs folder was removed and the interface was rewritten, cover protected reopening after reload, JSON download contents under all three policies, filters, hidden-action navigation, deletion, loading controls and a 390-pixel viewport without horizontal overflow. Synthetic services and an in-memory database were used. Native print output, a complete live-network browser run for the ML revision and full-duration browser timeout expiry were not established by those records. Backend deadlines and manual cancellation were checked separately. [1]

9 Machine learning evaluation

Dataset and evaluation method

career_profiles_dataset.csv contains 75 nonempty labelled text/role pairs, with 15 examples per class. The current evaluation normalises text case and whitespace, removes exact duplicates and stops on conflicting duplicate labels. The dated run found no duplicates, conflicts or screened train/test near-duplicate pairs at character similarity 0.9. Original source, collection method, consent, licence and annotation process remain unverified. The dataset must not be described as a representative sample of real professionals.

The dated repository evaluation uses TF-IDF unigrams/bigrams, English stop words, at most 4,000 features and sublinear term frequency, followed by logistic regression with C = 1.5, max_iter = 1,000 and random_state = 42. A stratified 80/20 holdout with seed 42 gives 60 training and 15 test examples. Five shuffled stratified folds also use seed 42, with the complete pipeline fitted within each fold. The latest record is dated 11 October 2026 in Asia/Colombo. Evaluation does not replace the runtime model unless explicitly requested. [1, 3, 4, 8]

Measure	Recorded result
Fold accuracies	100%, 100%, 100%, 100%, 93.33%
Mean five-fold accuracy	98.67%
Fold standard deviation	2.67 percentage points
Holdout accuracy	100% (15 of 15)
Holdout macro precision / recall / F1	1.000 / 1.000 / 1.000
Per-role holdout support	3 examples; all correctly classified
Exact duplicate rows / split overlap	0 / 0 (overlap is counted by split index; the safeguard is normalised deduplication before splitting)

Holdout confusion matrix

Rows are actual classes and columns are predicted classes. AE = AI Engineer, DS = Data Scientist, EN = Entrepreneur, RE = Researcher, SE = Software Engineer.

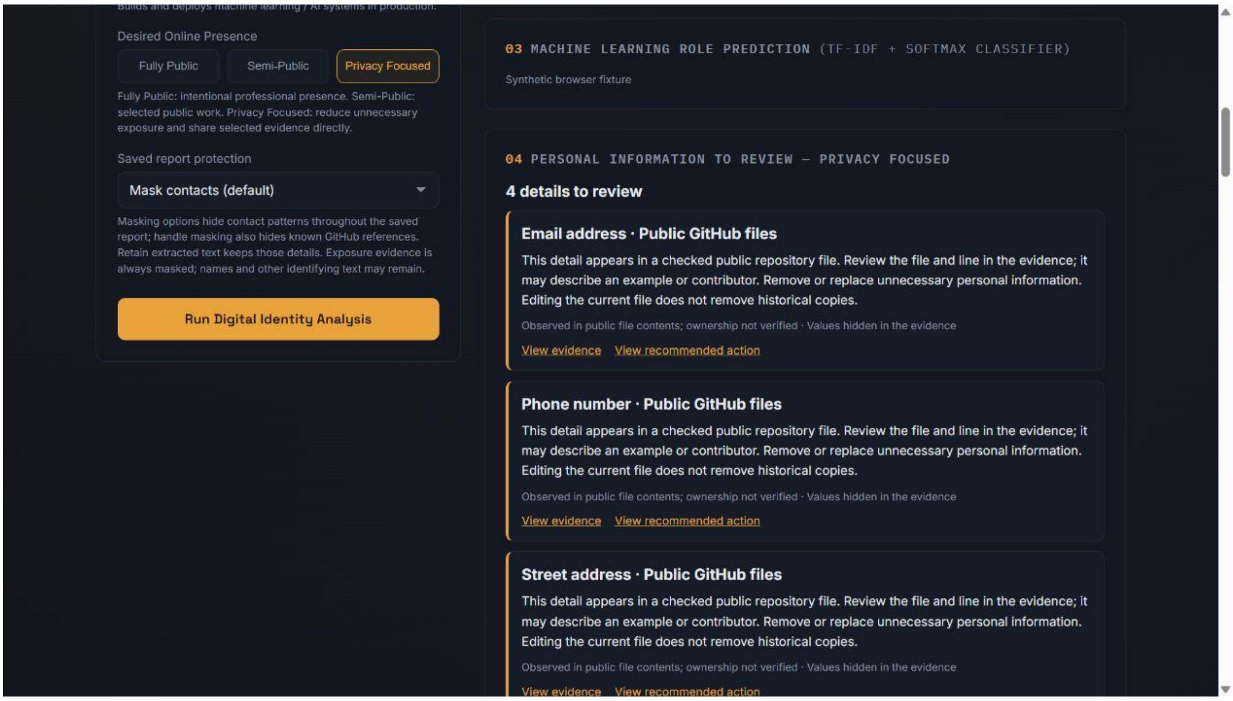
Actual	AE	DS	EN	RE	SE
AE	3	0	0	0	0
DS	0	3	0	0	0
EN	0	0	3	0	0
RE	0	0	0	3	0
SE	0	0	0	0	3

Interpretation

The changed cross-validation procedure explains why this result differs from the earlier unshuffled 97.33% baseline; the change is not evidence of improved real-world model quality. Exact and character-similarity checks cannot rule out semantic overlap, common templates or shared authors. No independent test population, probability calibration study or fairness evaluation is available. Zero-vocabulary abstention is implemented, but low-information text with matched vocabulary can still receive a role ranking. These limits should accompany the reported metrics.

10 User interface evidence

Figure 2 shows a synthetic browser verification image captured during the public repository-file privacy stage. It illustrates masked file evidence and scan coverage. It predates the history/export, filtering and de4d93f interface changes and is not presented as the latest complete interface. The current review controls and recorded checks are described below. No real user profile appears in this fixture. [1]



Builds and deploys machine learning / AI systems in production.

Desired Online Presence

Fully Public: intentional professional presence. Semi-Public: selected public work. Privacy Focused: reduce unnecessary exposure and share selected evidence directly.

Saved report protection: Mask contacts (default)

Masking options hide contact patterns throughout the saved report; handle masking also hides known GitHub references. Retain extracted text keeps those details. Exposure evidence is always masked; names and other identifying text may remain.

03 MACHINE LEARNING ROLE PREDICTION (TF-IDF + SOFTMAX CLASSIFIER)

Synthetic browser fixture

04 PERSONAL INFORMATION TO REVIEW — PRIVACY FOCUSED

4 details to review

Email address · Public GitHub files

This detail appears in a checked public repository file. Review the file and line in the evidence; it may describe an example or contributor. Remove or replace unnecessary personal information. Editing the current file does not remove historical copies.

Observed in public file contents; ownership not verified · Values hidden in the evidence

Phone number · Public GitHub files

This detail appears in a checked public repository file. Review the file and line in the evidence; it may describe an example or contributor. Remove or replace unnecessary personal information. Editing the current file does not remove historical copies.

Observed in public file contents; ownership not verified · Values hidden in the evidence

Street address · Public GitHub files

This detail appears in a checked public repository file. Review the file and line in the evidence; it may describe an example or contributor. Remove or replace unnecessary personal information. Editing the current file does not remove historical copies.

Observed in public file contents; ownership not verified · Values hidden in the evidence

Figure 2 Historical synthetic file-privacy verification preview. The current implementation additionally provides protected JSON export, print and deletion of the displayed report, recommendation filters and a simplified explanation summary.

Current controls: the Saved reports panel is hidden at 59662cd, although the backend history endpoints remain. Export JSON downloads the displayed report, Print report / Save PDF restores filters after including all recommendations, and Delete saved report asks for confirmation.

Recommendation review: career/privacy/legacy category and priority controls retain original ranks. Privacy evidence cards link to recorded evidence and, where available, to the recommended action. Historical actions remain readable.

Explanation summary: since de4d93f the Explanation Summary section opens with simplified status cards for target role match, information provided, GitHub activity, AI role prediction and privacy exposure. The GitHub activity, role prediction and privacy cards appear only when the underlying evidence exists, and the full technical narrative is available in a collapsible section. Input controls carry (?) help tooltips that explain resume limits, resume publicity, GitHub retrieval, repository-file review, LinkedIn text, visibility levels and report protection.

11 Limitations and discussion

Evidence and language limitations

Dictionary coverage, aliases and approximate spelling matching can miss unfamiliar terminology or create false positives. Context boundaries reduce mistakes but do not provide complete language understanding. Cross-source precedence can suppress an explicit denial when another source claims the same skill. Planned learning earns limited score credit and must remain visibly distinct from demonstrated ability. Employment estimates use explicit totals or year ranges under recognised headings; month-level dates, career breaks and unusual formatting may remain unknown.

Benchmark and model limitations

Five fixed benchmarks simplify role variation and may favour particular technologies or public portfolios. Repository counts and primary languages are proxies rather than verified quality. Fuzzy parameters and contextual weights are hand-authored and have not been validated against expert judgements. The ML dataset is small, its provenance is unknown and probabilities are uncalibrated. TF-IDF vocabulary matches identify terms present in the input; they do not establish causal explanations or measured class contributions.

Coverage and reliability limitations

GitHub retrieval analyses one updated-time-sorted page of at most 100 public repositories, including forks in the fetched-page limit. Career portfolio metrics use non-forks; incomplete coverage suppresses affected advice. File privacy review samples up to five fetched repositories and root allowlisted files only. A skipped file or unavailable source is not a clean privacy finding. Rate limits, missing files and execution bounds can make coverage partial.

The backend deadline starts after upload reading and does not bound final SQLite persistence or the multipart parser's initial buffering. Abandoned Python workers may finish their current operation. Browser cancellation can occur before a server saves its report; no idempotency or background-job management exists. These safeguards improve responsiveness but do not demonstrate high-load throughput.

Future research and product work

The nine agreed improvements are complete. Further work includes expert-reviewed benchmarks, independently annotated extraction data, documented and licensed classification data, external evaluation, probability calibration, a validated ambiguity threshold and broader accessibility assessment. Backend report history is limited to the latest 50 entries and its interface panel is hidden; multi-user access control, job cancellation/idempotency and native print verification remain open. OCR, multilingual processing, private-account access and repository history review require separate designs.

No user study, recruitment-outcome study, production security audit or independent fairness assessment has been completed. The system should support reflective self-review and human judgement, with uncertainty visible beside its outputs. [1]

12 Ethical and social considerations

Privacy and user control

Visibility preference and report protection solve different problems. Fully Public, Semi-Public and Privacy Focused guide professional-sharing advice. mask_contacts, mask_contacts_and_handle and none control the saved/returned report independently. Common contact masking is pattern-based and cannot guarantee anonymity: names, identifying prose and unsupported formats can remain. The none option can retain supplied extracted text, but exposure evidence stays masked and raw repository-file downloads are never saved.

A privately submitted resume is application evidence, not proof of public exposure. A declared public resume is assessed on the user’s declaration. Supplied LinkedIn text does not establish its actual audience or settings. Public GitHub metadata/files establish the location of a possible exposure, but not whether the contact belongs to the account owner or is inappropriate. Contributor and example details can be false positives. The tool advises review and does not change external content or settings.

Bias accuracy and misuse

Benchmark technology choices, English vocabulary and GitHub portfolio proxies may disadvantage users with private work, limited connectivity, different professional traditions or underrepresented careers. Privacy-focused advice should support private portfolio development and selective sharing. Unsupported evidence should produce uncertainty, rather than a judgement of low ability. Predictions and ratios must not be used as automatic hiring filters or rankings of personal worth.

Security retention and responsibility

SQLite persistence creates a local record that users must manage. The interface can delete a saved row, but deletion does not erase exports, backups, external posts or historical copies. Editing a current README cannot remove old commits or third-party copies. Because the prototype lacks login and report ownership checks, it should remain a local demonstration until access control is designed and verified. Masking is not a replacement for authorisation.

The user remains responsible for deciding whether to correct a profile, develop a skill or remove unnecessary public information. Authors are responsible for accurate references and contribution claims. The assignment requires acknowledgement of AI-generated assistance. The acknowledgement records that assistance while retaining human responsibility for the submitted work. [2]

13 Conclusion and reproducibility

Conclusion

DIAR demonstrates a practical AI decision-support workflow for professional identity review. It combines structured benchmark knowledge, context-aware extraction, learned role classification, fuzzy evidence summaries and traceable rules. The resulting recommendations explain why an action is suggested and distinguish source limitations from established facts. Independent report protection and bounded public-file review extend privacy control beyond a simple visibility filter.

The reviewed implementation completes all nine agreed improvements and incorporates the work merged through PR #11. The latest recorded regression run passed 199 tests. The dated evaluation obtained 98.67% mean shuffled five-fold accuracy and 100% holdout accuracy on the small local dataset. These results establish a reproducible internal baseline while leaving real-profile generalisation, provenance, calibration and fairness unverified.

Reproducible project package

The project package consists of the prototype source, pinned dependency files, test suite, benchmark and skill dictionaries, classification dataset, training/evaluation script and dated evaluation records. The assignment also requires presentation slides, a recorded video and an individual contribution sheet. The final report describes the delivered implementation and its limitations. [1, 2]

Release baseline

The reference source is the DIAR GitHub repository at main commit 59662cd, checked on 17 October 2026. It incorporates implementation commit f635bd6 through merge commit e2e0440 and the later interface and documentation commits de4d93f, 6d7678f and 186ef77. Verification and evaluation dates are retained separately from this report revision date. Historical saved reports are not automatically rescored when code changes, so a report read back from storage remains a record of its original analysis.

Evaluation record

The dataset SHA-256 is 9182020e17af7b8537e37a532048b8d3b3a131186e948dedc6a0a6e49185e50f. The dated run used 75 unique profiles, 15 per class, a 60/15 holdout and five shuffled stratified folds with seed 42. The JSON record preserves split indices, a confusion matrix, per-class metrics, package versions and limitations. Re-evaluation updates the timestamp and should retain numerical results under the pinned configuration, subject to floating-point/platform variation. [8]

The verified environment records Python 3.12.10 and pinned FastAPI, AnyIO, Uvicorn, python-multipart 0.0.32, PyMuPDF, SQLAlchemy, Pydantic, requests, scikit-learn and joblib dependencies. requirements.txt lists direct runtime versions, requirements-dev.txt adds test packages and requirements-lock.txt captures resolved transitive dependencies. The workflow configuration is included, and GitHub Actions records successful Backend regression runs for e2e0440 and all later main commits through 59662cd. [1]

Acknowledgement of assistance

OpenAI Codex assisted with repository inspection, earlier test and evaluation execution, and report drafting and formatting. The group is responsible for reviewing technical accuracy and references. Individual contributions are recorded in Appendix A (Individual contributions) of this report.

14 References

[1] DIAR project group. DIAR source repository. https://github.com/BuwanjithKasunara/diar-project. Reviewed 17 October 2026 at main commit 59662cd, following merge commit e2e0440 (implementation commit f635bd6) and later commits de4d93f, 6d7678f and 186ef77. Evidence includes application modules, frontend/index.html, frontend/report-review.js, regression tests, pinned requirements, the CI workflow and its GitHub Actions run history; development verification records under docs/ were removed in 6d7678f and are cited as historical records at e2e0440.

[2] General Sir John Kotelawala Defence University. Essentials of Artificial Intelligence Group Assignment. Supplied assignment brief, pages 1 to 5. Final report requirements on page 4 and participation, references and academic-integrity requirements on page 5. Source: Essentials of AI_Group Assignment (1).pdf.

[3] scikit-learn developers. Linear Models, Logistic Regression. Official user guide. https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression. Accessed 9 October 2026. Supports the classifier methodology, not DIAR's measured accuracy.

[4] scikit-learn developers. Feature Extraction, Text Feature Extraction. Official user guide. https://scikit-learn.org/stable/modules/feature_extraction.html#text-feature-extraction. Accessed 9 October 2026. Supports TF-IDF terminology and method.

[5] FastAPI. Concurrency and async await. Official documentation. https://fastapi.tiangolo.com/async/. Accessed 9 October 2026. Background for separating blocking processing from asynchronous request handling.

[6] GitHub. REST API endpoints for repository contents. Official documentation. https://docs.github.com/en/rest/repos/contents. Accessed 9 October 2026. Background for public repository content retrieval; DIAR's narrower allowlist and budgets are project decisions.

[7] DIAR project group. DIAR Project Proposal. Supplied university project document. Provides the problem definition, target users, objectives, proposed AI techniques and intended architecture.

[8] DIAR project group. Model evaluation and dataset record, 11 October 2026. Repository path docs/testing/model-evaluation.json, restored in 59662cd; the companion model-evaluation.md was removed in 6d7678f and remains available at e2e0440. https://github.com/BuwanjithKasunara/diar-project/blob/59662cd1997a6339e75f5415dcd55d031e2cec23/docs/testing/model-evaluation.json. Records shuffled cross-validation, seeded holdout, dataset identity and duplicate screening; provenance and generalisation limits remain explicit.

Appendix A Individual contributions

This appendix gives one entry per member in the format of the Individual Contribution Sheet. Entries are derived only from the repository history at main commit 59662cd and its remote branches: commit authors and committers, pull-request authors and merges, co-author trailers, files changed and git line statistics. Line counts include tests, data and documentation. Development plans in the repository assigned commit authorship per improvement, so the git record shows recorded authorship rather than a measure of individual effort. Work that the repository cannot verify is marked [To be completed by member].

Identity mapping: BuwanjithKasunara and Buwanjith Kasunara (buwanjithkasunara@gmail.com, repository owner) correspond to B.K.B Kasunara, and Sithum Alwis and sda2003 (GitHub profile name Sithum Alwis) correspond to A.M.S.D Alwis. The accounts raveesha2002 and charya19 show no real names. They are provisionally mapped to V.A.R. Gimhani and H.G.C Pathumini from their initials and the planned responsibilities recorded in the repository; both members must confirm this mapping.

ESSENTIALS OF ARTIFICIAL INTELLIGENCE

INDIVIDUAL CONTRIBUTION SHEET

Student Name	H.G.C Pathumini	**Registration No.**	D/DBA/25/0030
Degree / Intake	42	**Group No.**	41
Project Title	AI-Based Digital Identity Analysis and Recommendation System (DIAR)		

Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
Saved-report protection	Separated saved-report protection from the visibility preference and displayed exposure evidence.	Python, FastAPI; pattern-based contact masking; pytest	Commit 64227f0, PR #5 (8 Oct 2026): privacy.py, main.py, tests/test_privacy.py, frontend/index.html
Improvement 7: failure recovery	Preserved the previous report on failure and isolated slow processing behind worker deadlines.	Python, FastAPI, AnyIO worker threads; pytest	Commit 71c4553, PR #7 (9 Oct 2026): main.py, ml_classifier.py, tests/test_request_recovery.py
Improvement 9: repository file privacy scan	Added the bounded public repository file scan with masked evidence and coverage records.	Python, GitHub REST API; pattern-based privacy detection; pytest	Commit 4099c76, PR #8 (9 Oct 2026): repository_privacy.py, tests/test_repository_file_privacy.py
Improvements 5 and 6: report history, export and navigation	Added saved-report history, protected JSON export and recommendation filters and navigation.	JavaScript, FastAPI; pytest	Commit c53cece, PR #10 (11 Oct 2026): frontend/report-review.js, frontend/index.html, tests/test_recommendations.py
Improvement 8: reproducible setup	Pinned dependencies, added the CI workflow and recorded the model evaluation.	GitHub Actions, pip lockfile; scikit-learn evaluation	Commit f635bd6, PR #11 (11 Oct 2026): .github/workflows/ci.yml, requirements files, train_ml_model.py, tests/conftest.py, docs/testing/model-evaluation.json

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature: charya **Date:** 17/10/2026

Evidence may include source code, model/notebook, diagram, dataset work, test results, screenshots, report section or presentation slides.

ESSENTIALS OF ARTIFICIAL INTELLIGENCE

INDIVIDUAL CONTRIBUTION SHEET

Student Name	V.A.R. Gimhani	**Registration No.**	D/BIT/24/0049
Degree / Intake	41	**Group No.**	41
Project Title	AI-Based Digital Identity Analysis and Recommendation System (DIAR)		

Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
Source-specific visibility recommendations	Added visibility rules and source-specific privacy advice, then corrected omitted findings and source-coverage advice.	Python; rule-based reasoning; pattern-based privacy assessment; pytest	Commits 398dce9 and 2cb1f22, PR #4 (8 Oct 2026): privacy_assessment.py, alignment_engine.py, recommendation_engine.py, tests/test_privacy_evidence.py
Privacy findings interface	Made privacy findings easier to review in the interface.	HTML, CSS, JavaScript	Commit 2a6c513, PR #5 (8 Oct 2026): frontend/index.html
Improvements 1 to 3: validation and career evidence	Validated inputs and uploads, handled insufficient evidence and unknown experience, and improved skill-context extraction and alias deduplication.	Python, PyMuPDF; dictionary and regex language processing with negation and planning context; pytest	Commit 91f9962, PR #6 (8 Oct 2026): extraction.py, main.py, identity_construction.py, tests/test_analysis_foundation.py
Improvement 4: ML evidence	Clarified prediction uncertainty, insufficient-evidence results and TF-IDF vocabulary explanations.	scikit-learn TF-IDF and logistic regression; explainable AI; pytest	Commit 097169a, PR #9 (11 Oct 2026): ml_classifier.py, explainable_ai.py, tests/test_ml_evidence.py

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature: Raveesha **Date:** 17/10/2026

Evidence may include source code, model/notebook, diagram, dataset work, test results, screenshots, report section or presentation slides.

ESSENTIALS OF ARTIFICIAL INTELLIGENCE

INDIVIDUAL CONTRIBUTION SHEET

Student Name	A.M.S.D Alwis	**Registration No.**	D/BCS/25/0020
Degree / Intake	42	**Group No.**	41
Project Title	AI-Based Digital Identity Analysis and Recommendation System (DIAR)		

Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
Recommendation labels	Labelled all recommendation actions and preserved original skill names.	Python; rule-based reasoning; pytest	Commit 3430c4a, PR #1 (8 Oct 2026): recommendation_engine.py, tests/test_recommendations.py
GitHub evidence handling	Withheld portfolio judgements when GitHub evidence is unavailable or incomplete.	Python, GitHub REST API; rule-based reasoning over evidence coverage; pytest	Commit 17eec93, PR #2 (8 Oct 2026): extraction.py, alignment_engine.py, tests/test_github_evidence.py, frontend/index.html
Privacy exposure evidence foundation	Added the exposure-evidence assessment foundation and privacy realignment documents.	Python; pattern-based privacy detection; pytest	Commit 08d0b56, PR #3 (8 Oct 2026): privacy_assessment.py, tests/test_privacy_evidence.py
Pull-request integration	Opened pull requests #1 to #11 and merged each into main.	GitHub pull requests	11 merge commits from 84f6397 (PR #1, 8 Oct 2026) to e2e0440 (PR #11, 11 Oct 2026)
Dataset and annotation workflow (not merged into main)	Evidence-scoring redesign, central profile dataset pipeline, annotation pilot and adjudication workflow.	Python data scripts, JSON schemas; pytest	Commits 9af881f, 19eda03, 7c5aa54, 6ef4eb3 and d88452e (4 to 5 Oct 2026; d88452e co-authored with Copilot), +8,452 / −1,744 lines
Report and guide documents	Added a final report and presentation template and branch-improvement notes, both later removed from main.	Markdown	Commits 2aa8997 and ddbfe28 (7 Oct 2026)

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature: **Date:** 17/10/2026

Evidence may include source code, model/notebook, diagram, dataset work, test results, screenshots, report section or presentation slides.

ESSENTIALS OF ARTIFICIAL INTELLIGENCE

INDIVIDUAL CONTRIBUTION SHEET

Student Name	B.K.B Kasunara	**Registration No.**	D/BIT/24/0061
Degree / Intake	41	**Group No.**	41
Project Title	AI-Based Digital Identity Analysis and Recommendation System (DIAR)		

Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
Initial DIAR prototype & System Architecture	Created the initial backend and interface: FastAPI application, SQLite persistence, schemas, extraction, identity construction, alignment, fuzzy logic, recommendation and explanation modules, benchmark and skill dictionaries. Designed system architecture in modular, multi-tier intelligence pipeline	Python, FastAPI, SQLAlchemy/SQLite, JavaScript; dictionary-based language processing, knowledge representation, fuzzy logic, rule-based reasoning	Commit f2ef973 (23 Aug 2026), 17 files
ML role classifier and scoring	Added the role classifier, training script and dataset and labelled career-profile dataset, context-aware skill extraction, multi-factor scoring and a pipeline test suite, XAI explanations for ML predictions, and automated pipeline test suite	scikit-learn TF-IDF & logistic regression; contextual NLP extraction; pytest; FastAPI	Commit ceff7ba (7 Oct 2026): ml_classifier.py, train_ml_model.py, career_profiles_dataset.csv, tests/test_pipeline.py
Interface rewrite	Added simplified explanation summary cards (Role Match, Profile Completeness, GitHub Activity, AI Role Prediction, Privacy Alerts) and interactive (?) help tooltips; streamlined recommendation navigation by removing redundant shortcuts.	HTML, CSS, JavaScript; explainable-AI presentation	Commit de4d93f (16 Oct 2026): frontend/index.html (+1,694 lines), frontend/report-review.js
Release clean-up	Cleaned up obsolete documentation, restored and verified the formal ML model evaluation record, refined UI display, and updated the README.	Git, Markdown, CSS	Commits 6d7678f (16 Oct 2026), 186ef77 and 59662cd (17 Oct 2026): docs/testing/model-evaluation.json, README.md, frontend/index.html

Repository setup and maintenance	Added README virtual-environment setup instructions (.venv), .gitignore maintenance, and comprehensive project documentation.	Git, GitHub	Commits dd690b7, ac25edc, 96a205b, 89f895e, 8700776, 2f5d00a and f1bf3f9 (7 Oct 2026)

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature: ________________ **Date:** 17/10/2026

Evidence may include source code, model/notebook, diagram, dataset work, test results, screenshots, report section or presentation slides.